



Computer Hardware Practical Guide

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System Unit Kya Hai?

System unit aik metallic ya plastic ka casing (case) hai jo computer ke andar ke tamaam main parts ko mehfooz (protect) krta hai.



Main Purpose:

Yeh sabhi internal components ko aapas mein connect karta hai aur external damage se bachata hai.



Motherboard



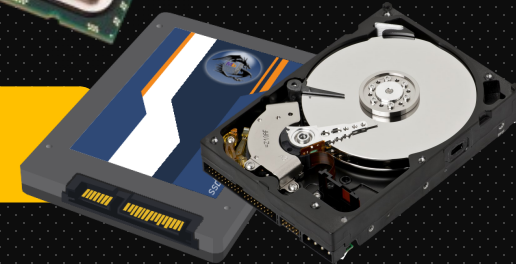
Power Supply Unit



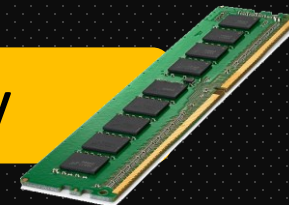
Central Processing Unit



Storage Devices HDD, SSD



RAM(Random Access Memory)

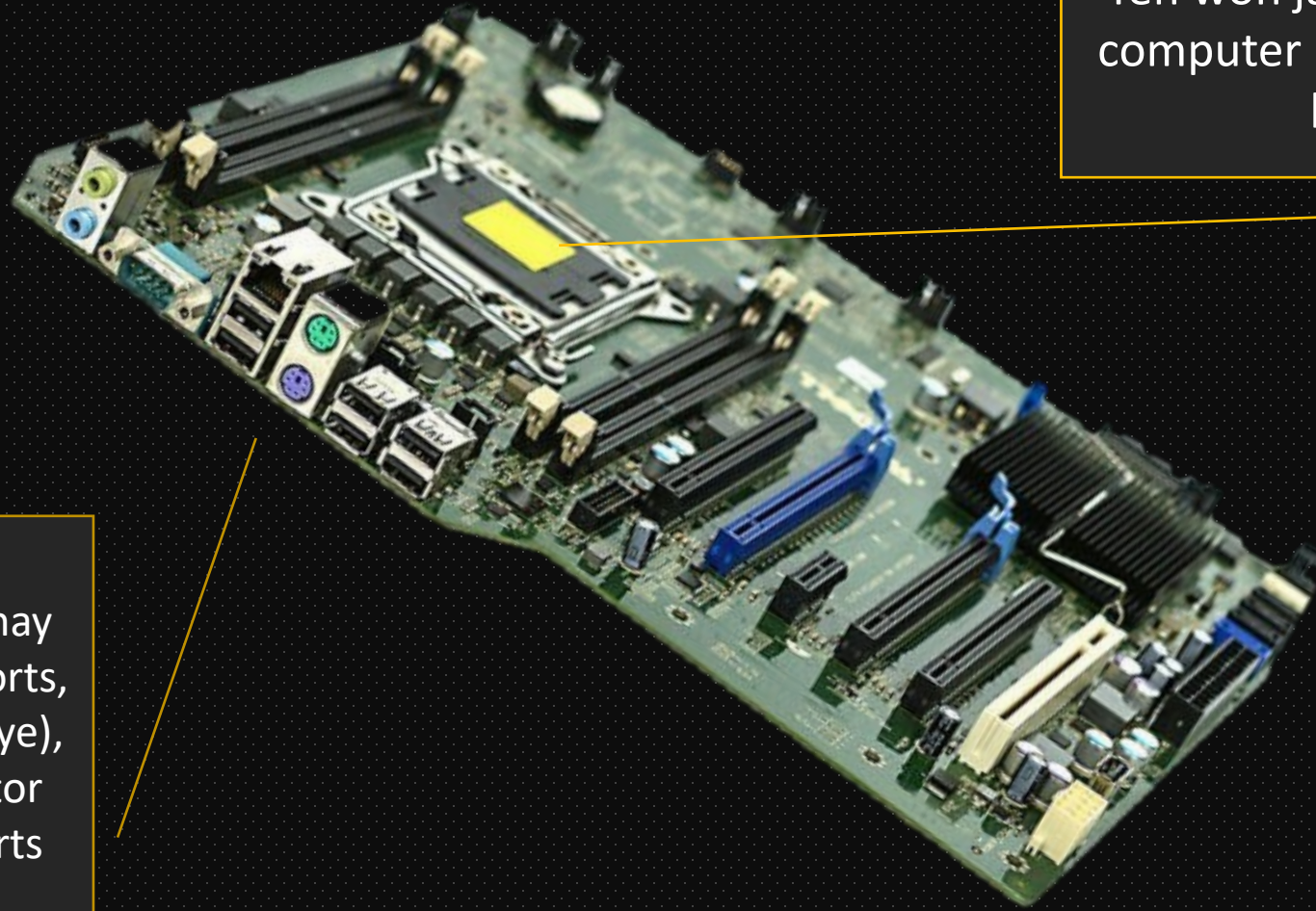


**System Unit ke
Main Parts**

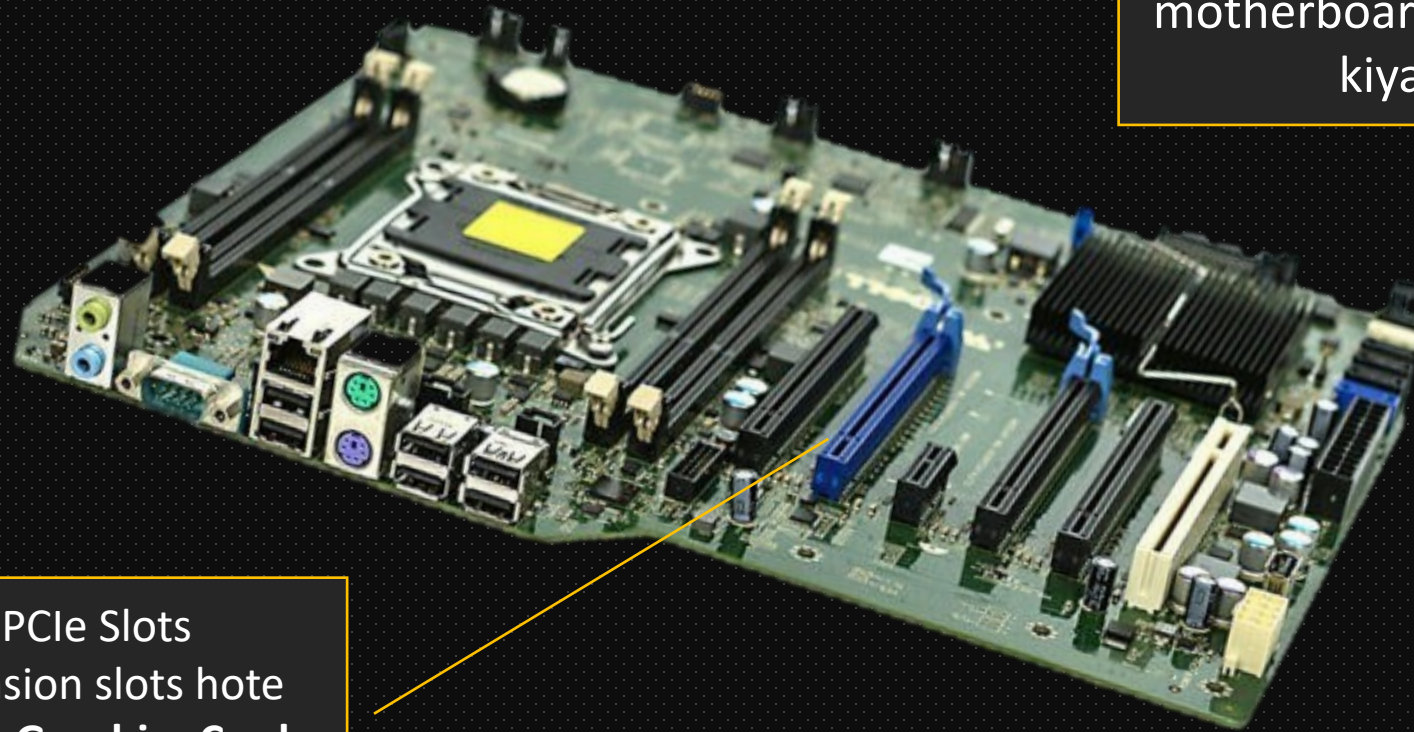
Motherboard Parts

CPU Socket:
Yeh woh jagah hoti hai jahan
computer ka **Processor (CPU)**
lagta hai.

Back Panel / Ports
Motherboard ke peechay
wale hisse mein USB ports,
LAN port (internet ke liye),
audio jacks, aur monitor
ke liye HDMI/VGA ports
hote hain.



Motherboard Parts



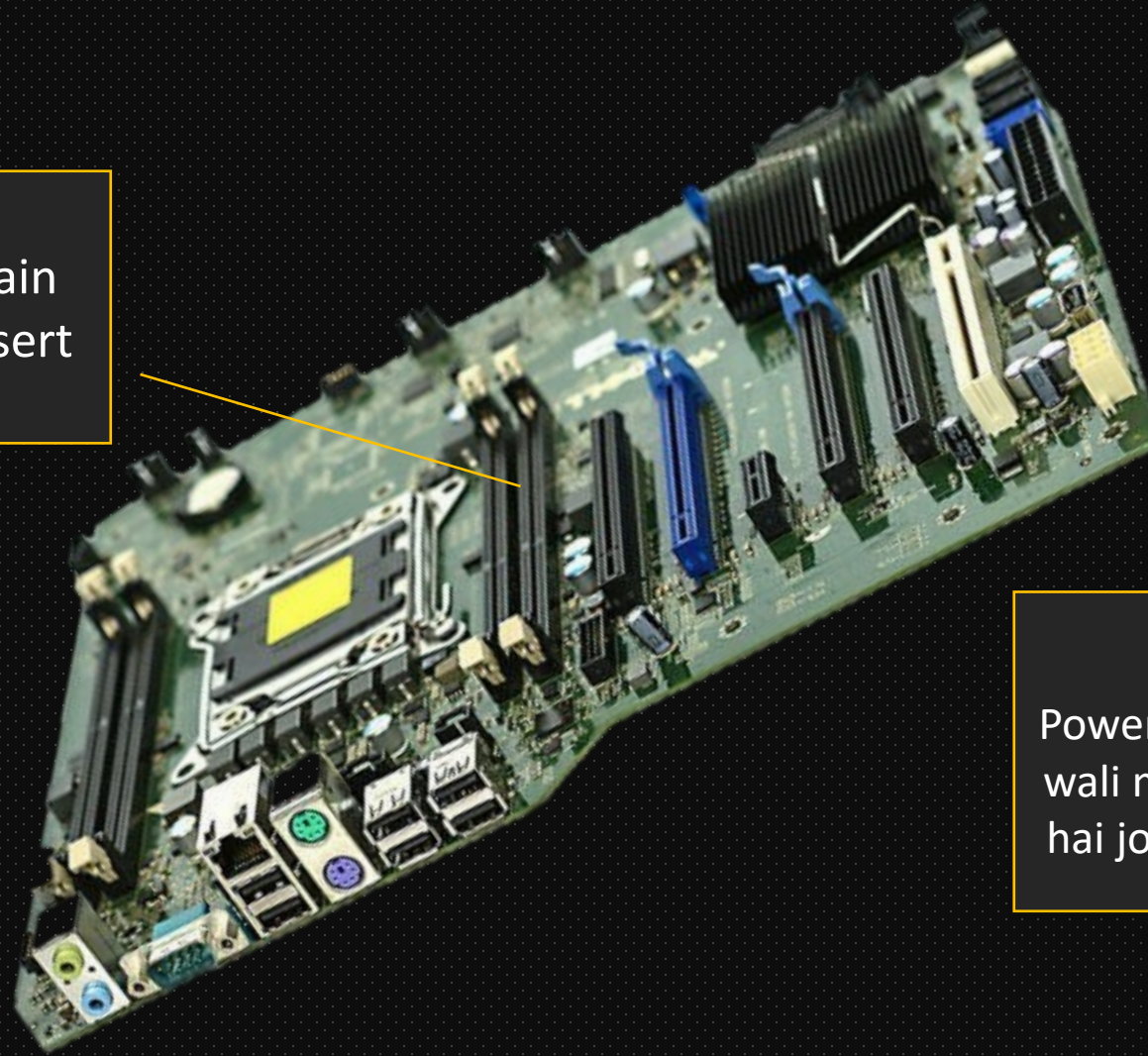
SATA Ports
In ports ke zariye **Hard Disk (HDD)** aur **SSD** ko motherboard ke sath connect kiya jata hai.

PCI / PCIe Slots
Yeh expansion slots hote hain jahan **Graphics Card (GPU)** ya network card lagaye jate hain.

Motherboard Parts

RAM Slots

Yeh lambay slots hote hain jahan **RAM** (Memory) insert ki jati hai.



Power Connectors

Power Supply Unit (SMPS) se aane wali main power cable yahan lagti hai jo pure board ko bijli deti hai..



ATX (Advanced Technology Extended):

Yeh standard aur **sab se zyada istemal hone wala** bara motherboard hota hai. Yeh aam gaming PCs ya office PCs mein lagta hai kyunki isme bohot saare slots (RAM, Graphics card, PCI) hote hain.



Micro-ATX:

Yeh ATX se **thora chota** hota hai.
Yeh darmiyani (medium) size ke computer cases
mein aata hai. Isme bhi achi khasisi cheezein lag
jaati hain lekin size compact hota hai.



Mini-ATX:

Yeh **bohot chota** motherboard hota hai.
Yeh chote ya portable PCs (jaise chote office boxes ya mini gaming setups) ke liye use hota hai, jahan space kam ho.



E-ATX (Extended ATX):

Yeh **sab se bara** motherboard hota hai. Yeh heavy-duty computers, servers, ya high-end gaming PCs ke liye use hota hai jisme bohot saari advanced cheezein lagani hon.

Power Supply Unit

Power Supply Unit (PSU) ya SMPS (Switched-Mode Power Supply) computer ka woh ahem hissa hai jo wall socket se aane wali high voltage alternating current (AC) ko direct current (DC) mein tabdeel karta hai, jo computer ke andar ke parts (jaise Motherboard, Processor, Hard Disk waghera) ko chalne ke liye zaroori hai.



Power Supply Unit

Main Kaam (Function)

Yeh ghar ki bijli (AC) ko low voltage DC mein convert karta hai taake computer ke sensitive parts kharab na hon.

Fan (Cooling):

Iske andar aik fan laga hota hai jo PSU ke andar ki garmi (heat) ko bahar nikalta hai aur system ko thanda rakhta hai.

Cables & Connectors:

Isme se mukhtalif rang ki cables nikalti hain jo motherboard, processor, hard disk, aur optical drives ko power supply karti hain.

Wattage:

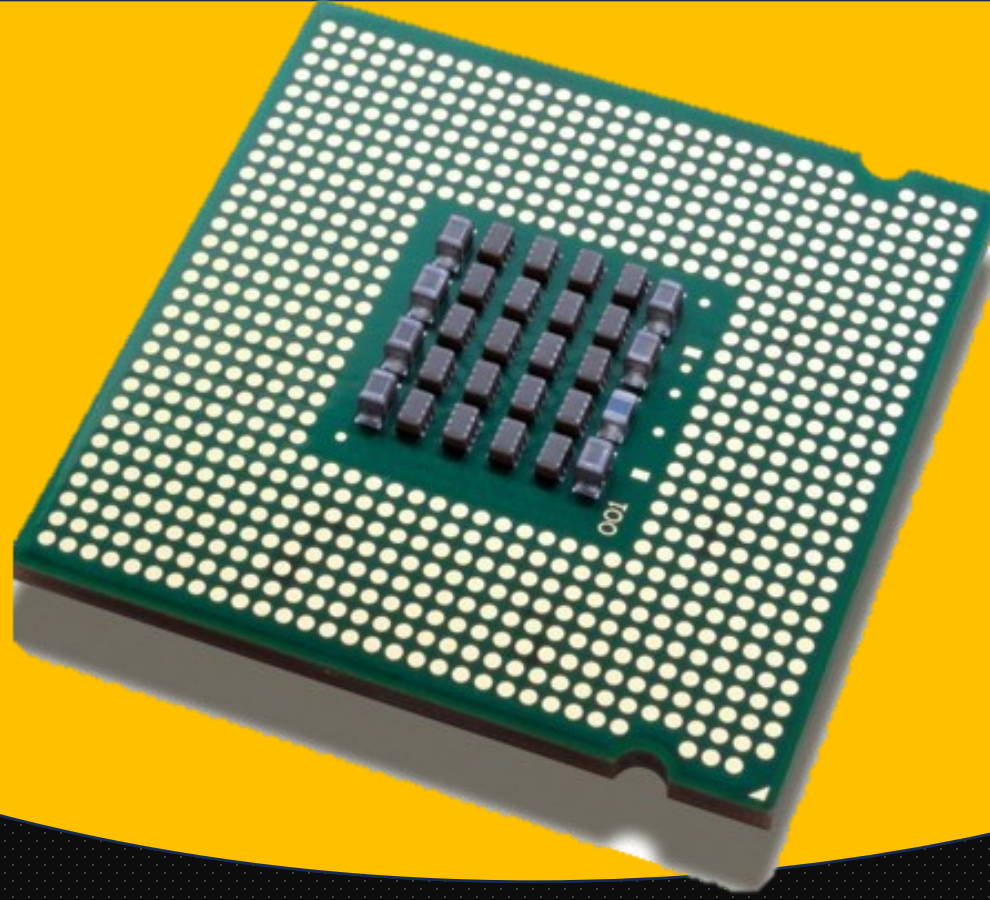
Power Supply alag-alag power (wattage) ki aati hai (jaise 350W, 500W, 700W) jo computer ke load ke hisab se lagai jati hai.





Core Definition

CPU computer ka **main processor** hai jo computer ke tamaam operations ko execute aur control karta hai. Ise computer ka **brain** is liye kehte hain kyunki software ki har instruction ki execution hoti hai.



Working

Jab aap koi command dete hain (jaise mouse click ya keyboard se typing), toh CPU us instruction ko fetch karta hai, decode karta hai (samajhta hai), aur phir execute kar ke screen par result hai.

Speed & Cores

CPU ki speed **GHz (Gigahertz)** mein measure hoti hai. Jitni zyada GHz ya cores honge (jaise Dual Core, Quad Core, i3, i5), CPU utni hi tezi se mukhtalif tasks aik sath perform ga.

Cooling System

CPU chalte waqt bohot ziyada garam (heat up) hota hai. Isliye iske ooper aik **Heatsink aur Cooling Fan** lagaya jata hai jo iski garmi ko absorb karke bahar nikalta hai, warna CPU over-heat ho kar band ho sakta hai.

Storage Devices Kya Hain?

Storage device computer ka woh ahem hissa hai jahan hum apna sara data (jaise Windows, software, movies, pictures, aur documents) **permanent** taur par save karte hain.

Computer band (shutdown) hone ke baad bhi iska data khatam nahi hota (non-volatile memory).



HDD (Hard Disk Drive)

Yeh purani aur traditional storage hai. Iske andar ghoomnay wali magnetic plates (disks) hoti hain jo data read/write karti hain. Yeh sasti milti hai aur capacity bohot ziyada hoti hai (jaise 500GB ya 1TB), yeh SSD ke muqablay mein slow hoti hai aur isse halki si awaz bhi aati hai.



3.5-Inch HDD (Desktop Hard Drive)



2.5-Inch HDD (Laptop Hard Drive)



SSD (Solid State Drive)

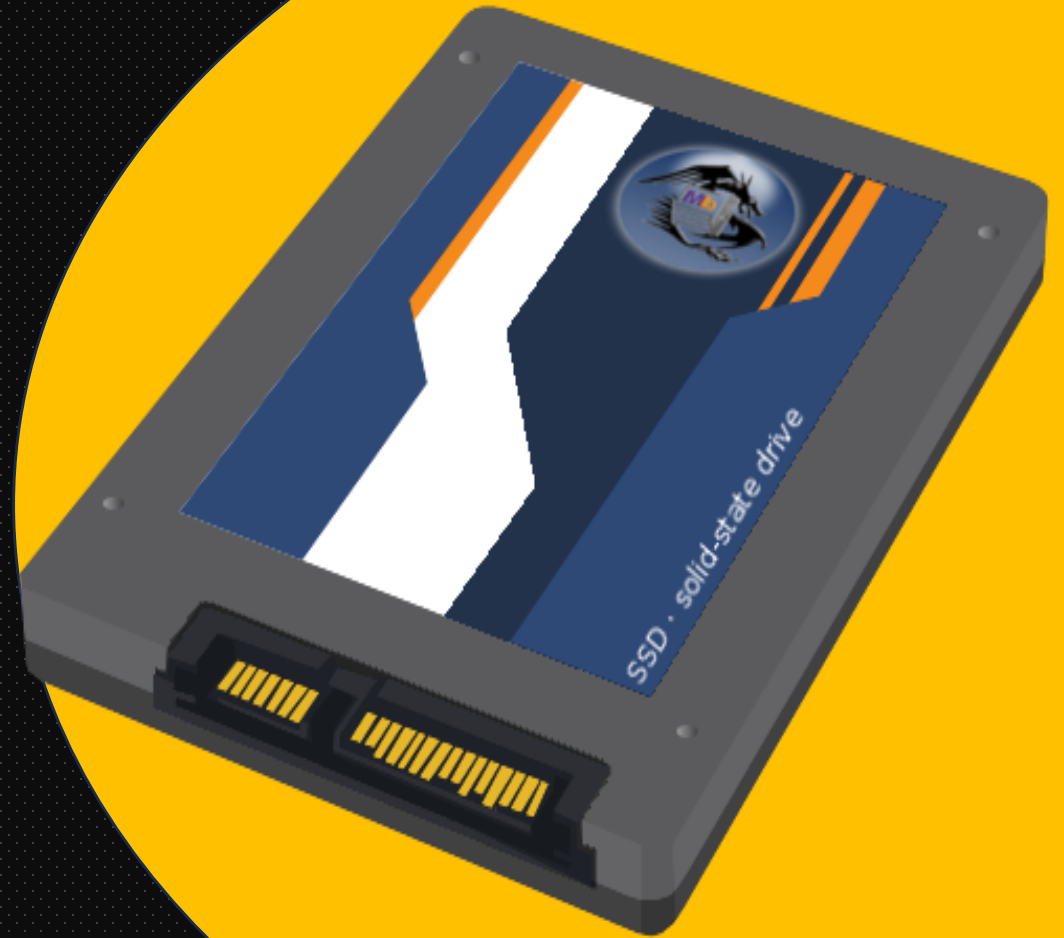
Yeh modern aur bohot fast storage device hai. Iske andar koi ghoomnay wali plate nahi hoti, balkay flash memory chips hoti hain (jaise mobile ka memory card ya USB). Is par Windows chalane se computer bohot tezi se on hota hai aur software foran hain.



2.5-Inch SATA SSD



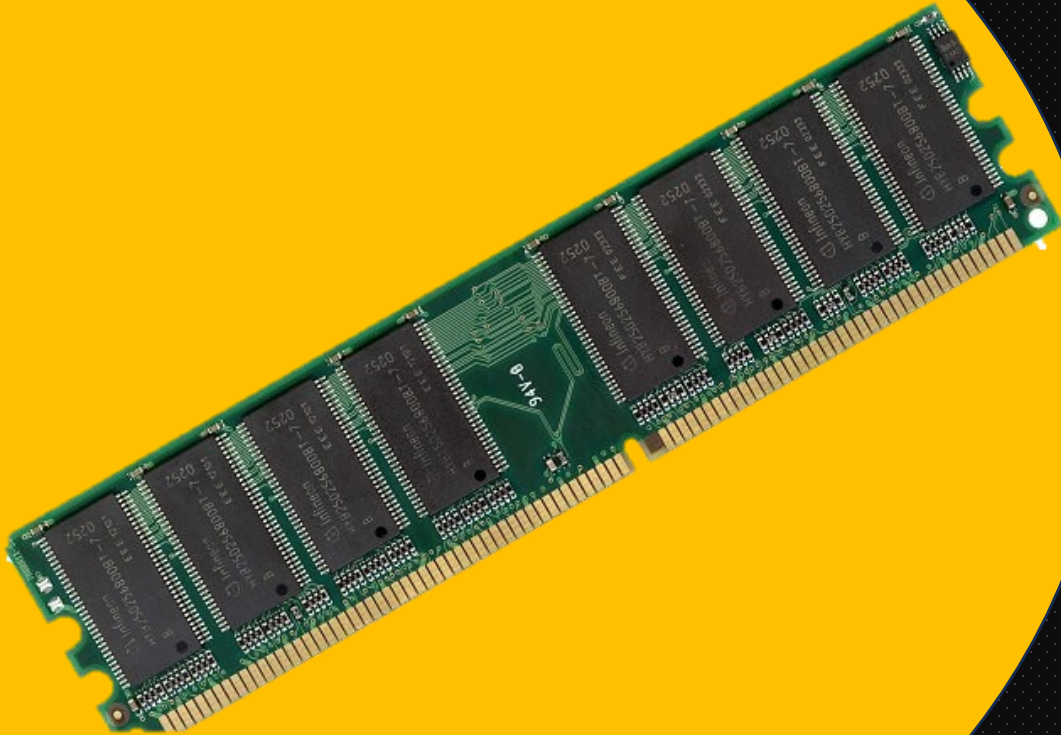
M.2 SATA SSD



RAM Kya Hai?

RAM ka matlab **Random Access Memory** hai.
Yeh computer ki **temporary aur primary memory** hoti hai.

Isse **volatile memory** bhi kehte hain, iska matlab yeh hai ke jab tak computer on hai, data RAM mein rahega; jaise hi computer band (shutdown) hoga, RAM ka sara data automatically delete ho jayega.

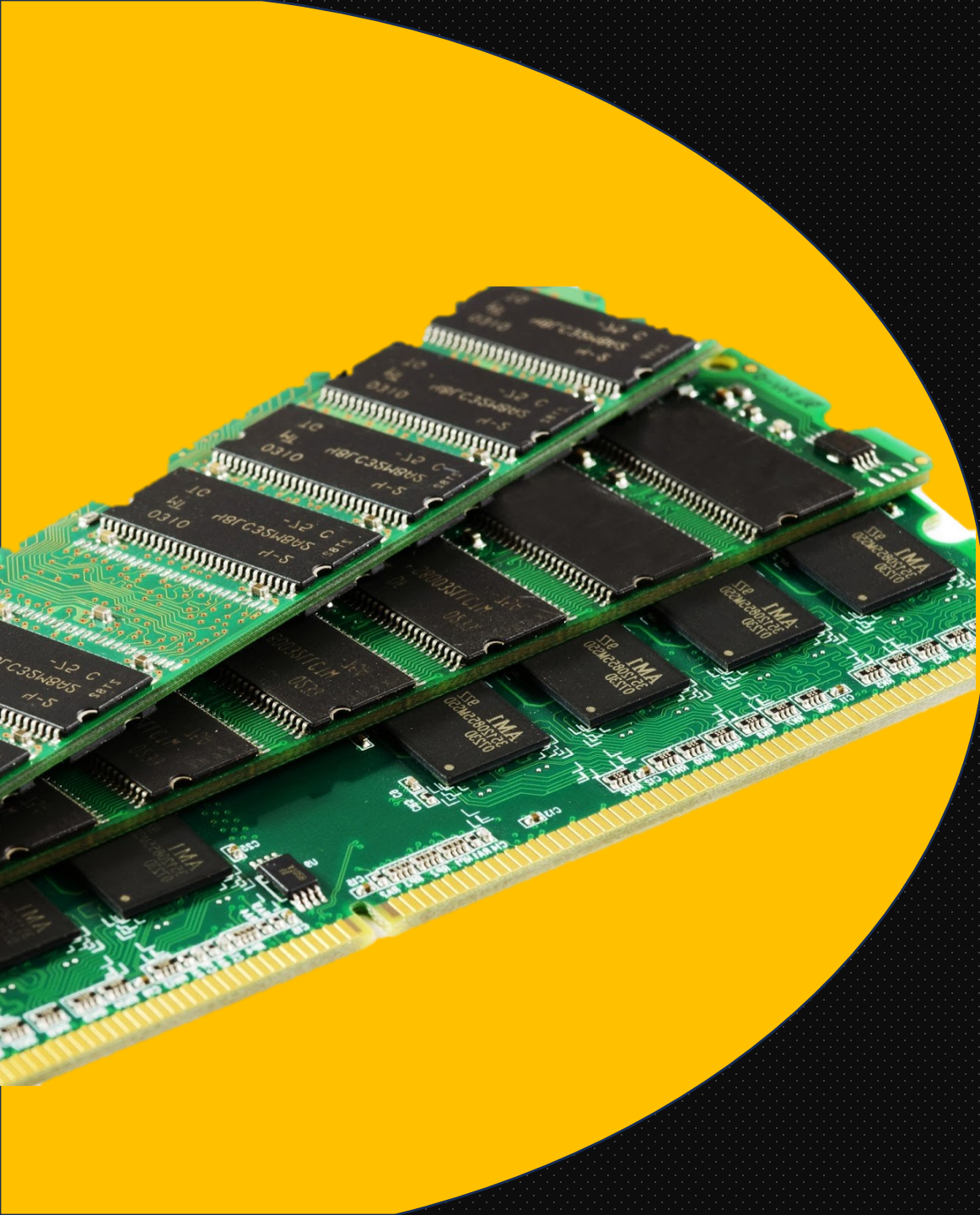


RAM Kaam Kaise Karti Hai?

Jab aap computer mein koi app, game, ya software open karte hain, toh computer us software ko apni permanent storage (Hard Disk ya SSD) se nikal kar **RAM** mein load kar deta hai.

Processor (CPU) phir baar-baar hard disk tak jane ke bajaye **direct RAM se data read aur write karta hai**, kyunki RAM hard disk ke muqablay mein hazaron guna tez hoti hai.

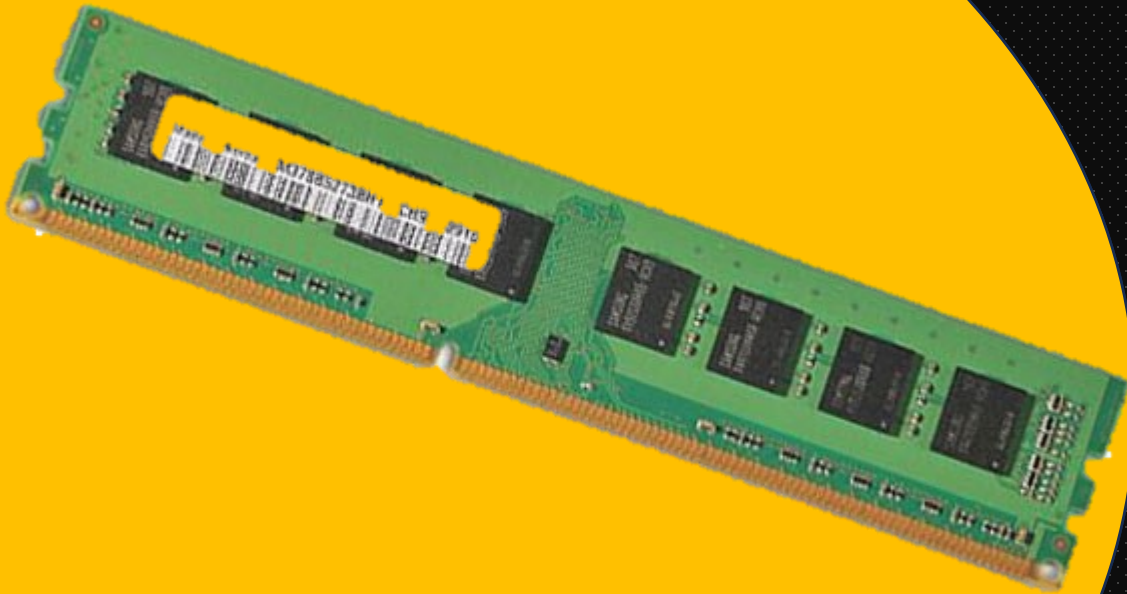
Multitasking: Jitni bari RAM hogi (jaise 8GB ya 16GB), computer aik waqt mein utne hi ziyada heavy programs baghair slow hue chala sake ga. Agar RAM choti ho aur aap bohot saari apps khol lein, toh computer hang ya slow ho jata hai.



Main Generations

DDR3

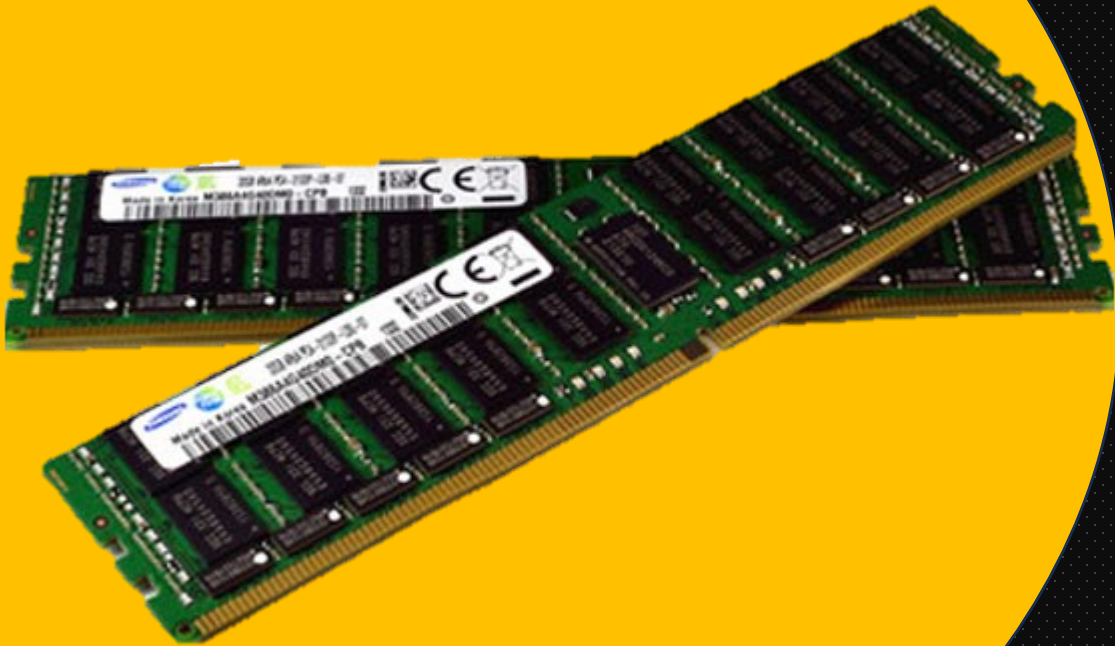
Yeh purane computers mein use hoti thi, ab yeh purani ho chuki hai.



Main Generations

DDR4

Yeh abhi zyadatar standard aur aam PCs mein use ho rahi hai, yeh kafi fast aur stable hoti hai.

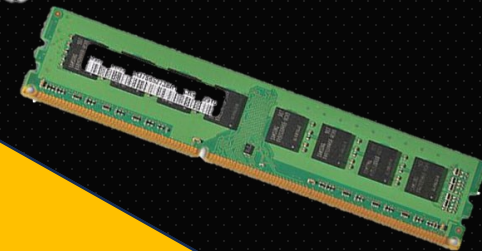
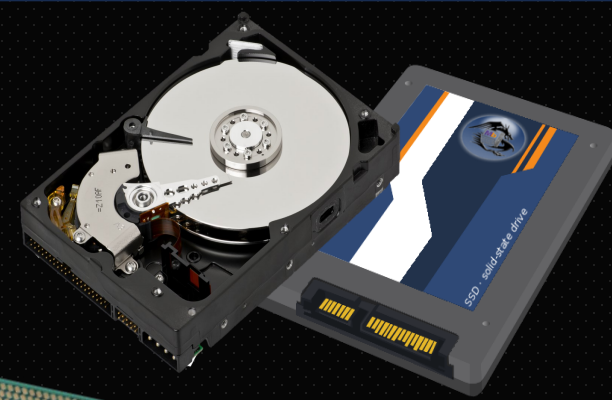


Main Generations

DDR5

Yeh sab se latest generation hai jo high-end aur modern gaming computers mein aati hai, aur yeh purani generations ke muqablay mein bohot hi zyada fast hoti hai.





Topic
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